Multimodal methods and AdaDiag in the format of the TVDiag paper: Tables 4 and 5

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Baselines and *TVDiag*: the paper's own numbers. AdaDiag: this repository, on the released datasets $\mathcal{A}$ (GAIA), $\mathcal{C}$ (Sock Shop) and $\mathcal{D}$ (Hotel Reservation). AdaDiag (best) is the AdaDiag configuration with the highest mean of HR@1 and F1 over $\mathcal{A}$, $\mathcal{C}$ and $\mathcal{D}$ among the 5 run on fresh seeds; the choice was made after the runs. Against *TVDiag* re-run under identical conditions it is not significantly different (README §6.19), so the gaps to the published *TVDiag* also reflect the re-implementation (README §6.1, §6.9).

Table 4. Root Cause Localization Comparison (multimodal methods)

Dataset	Method	p-value	*HR*@1	⇑ *HR*@1	*HR*@3	⇑ *HR*@3	*Avg*@3	⇑ *Avg*@3	*MRR*@3	⇑ *MRR*@3
$\mathcal{A}$	Eadro	1.6e-03	0.304	149.671%	0.592	53.547%	0.473	78.436%	0.437	89.245%
	DiagFusion	3.1e-02	0.404	87.871%	0.801	13.483%	0.636	32.704%	0.586	41.126%
	RF-RCL	3.4e-02	0.573	32.461%	0.815	11.534%	0.719	17.385%	0.686	20.554%
	TVDiag	-	**0.759**	-	**0.909**	-	**0.844**	-	**0.827**	-
	AdaDiag (best)†	2.0E-01	0.749	1.293%	0.899	1.156%	0.836	0.957%	0.818	1.158%
$\mathcal{B}$	Eadro	6.3e-04	0.226	295.575%	0.478	96.862%	0.367	151.771%	0.339	170.501%
	DiagFusion	4.3e-08	0.205	336.098%	0.273	244.689%	0.239	286.611%	0.229	300.437%
	RF-RCL	7.5e-03	0.635	40.787%	0.829	13.510%	0.739	25.034%	0.719	27.538%
	TVDiag	-	**0.894**	-	**0.941**	-	**0.924**	-	**0.917**	-
	AdaDiag: not run, dataset $\mathcal{B}$ (AIOps-22) is not released									
$\mathcal{C}$	Eadro	1.4E-04	0.191	351.832%	0.415	129.880%	0.317	188.013%	0.291	210.309%
	DiagFusion	9.0E-04	0.519	66.281%	0.710	34.366%	0.612	49.183%	0.597	51.256%
	RF-RCL	5.1E-03	0.622	38.746%	0.817	16.769%	0.715	27.692%	0.701	28.816%
	TVDiag	-	0.863	-	**0.954**	-	0.913	-	0.903	-
	AdaDiag (best)†	4.7E-02	**0.877**	−1.616%	0.949	0.575%	**0.919**	−0.647%	**0.910**	−0.765%
$\mathcal{D}$	Eadro	2.6E-03	0.244	236.475%	0.590	56.441%	0.419	109.069%	0.389	122.622%
	DiagFusion	4.5E-03	0.513	60.039%	0.744	24.059%	0.645	35.814%	0.618	40.129%
	RF-RCL	4.6E-02	0.744	10.349%	0.872	5.849%	0.804	8.955%	0.795	8.931%
	TVDiag	-	0.821	-	**0.923**	-	**0.876**	-	**0.866**	-
	AdaDiag (best)†	2.1E-01	**0.826**	−0.562%	0.915	0.832%	0.872	0.482%	0.864	0.269%

⇑ indicates the improvement rate of *TVDiag* compared to other methods, as in the paper (negative: the row is above *TVDiag*). The best result in each metric is shown in bold. Rows without † are copied from the paper (Xie et al., *TVDiag*, ACM TOSEM 35(2), Art. 40, 2026, Table 4); the single-modality baselines are omitted. †**AdaDiag (best)**: AdaDiag-DL (weights learned from the diagnosis losses, README §6.17) with Eq. 13's learnable loss weights (§6.19), mean over 5 seeds (5–9); p-value: two-sided one-sample t-test of the seeds against *TVDiag*'s published HR@1.

Table 5. Failure Type Identification Comparison (multimodal methods)

Dataset	Method	p-value	*Precision*	⇑ *Precision*	*Recall*	⇑ *Recall*	F1-score	⇑ F1-score
$\mathcal{A}$	DiagFusion	3.3E-04	0.852	8.568%	0.838	13.126%	0.840	11.429%
	Medicine	4.1E-06	0.669	38.266%	0.667	42.129%	0.648	44.444%
	DT-FTI	9.4E-03	0.889	4.049%	0.892	6.278%	0.890	5.169%
	SVM-FTI	9.9E-02	0.883	4.757%	0.935	1.390%	0.908	3.084%
	LGB-FTI	5.2E-02	0.892	3.700%	0.928	2.155%	0.907	3.197%
	TVDiag	-	0.925	-	0.948	-	0.936	-
	AdaDiag (best)†	2.5E-01	**0.930**	−0.575%	**0.953**	−0.495%	**0.941**	−0.573%
$\mathcal{B}$	DiagFusion	3.4E-04	0.533	65.478%	0.489	78.119%	0.505	73.465%
	Medicine	4.7E-04	0.307	187.296%	0.244	256.967%	0.244	259.016%
	DT-FTI	5.0E-05	0.448	96.875%	0.420	107.381%	0.422	107.583%
	SVM-FTI	3.6E-03	0.807	9.294%	0.784	11.097%	0.774	13.178%
	LGB-FTI	7.0E-04	0.604	46.026%	0.580	50.172%	0.556	57.554%
	TVDiag	-	**0.882**	-	**0.871**	-	**0.876**	-
	AdaDiag: not run, dataset $\mathcal{B}$ (AIOps-22) is not released							
$\mathcal{C}$	DiagFusion	7.9E-05	0.338	92.012%	0.332	87.349%	0.328	93.902%
	Medicine	5.0E-04	0.497	30.584%	0.469	32.623%	0.454	40.088%
	DT-FTI	6.4E-05	0.376	72.606%	0.365	70.411%	0.366	73.770%
	SVM-FTI	3.2E-03	0.384	69.010%	0.315	97.460%	0.282	125.532%
	LGB-FTI	9.7E-04	0.442	46.833%	0.390	59.487%	0.388	63.918%
	TVDiag	-	**0.649**	-	**0.622**	-	**0.636**	-
	AdaDiag (best)†	1.3E-02	0.629	3.148%	0.575	8.154%	0.601	5.859%
$\mathcal{D}$	DiagFusion	3.2E-05	0.618	35.275%	0.603	33.997%	0.600	37.000%
	Medicine	1.9E-02	0.782	6.905%	0.725	11.448%	0.728	12.912%
	DT-FTI	5.2E-03	0.676	23.669%	0.603	33.997%	0.598	37.458%
	SVM-FTI	9.5E-03	0.488	71.311%	0.628	28.662%	0.540	52.222%
	LGB-FTI	4.1E-02	0.809	3.337%	0.769	5.072%	0.764	7.592%
	TVDiag	-	0.836	-	0.808	-	0.822	-
	AdaDiag (best)†	1.8E-01	**0.852**	−1.930%	**0.823**	−1.832%	**0.837**	−1.848%

⇑ indicates the improvement rate of *TVDiag* compared to other methods, as in the paper (negative: the row is above *TVDiag*). The best result in each metric is shown in bold. Rows without † are copied from the paper (Xie et al., *TVDiag*, ACM TOSEM 35(2), Art. 40, 2026, Table 5); the single-modality baselines are omitted. †**AdaDiag (best)**: AdaDiag-DL (weights learned from the diagnosis losses, README §6.17) with Eq. 13's learnable loss weights (§6.19), mean over 5 seeds (5–9); p-value: two-sided one-sample t-test of the seeds against *TVDiag*'s published F1-score.